

FULL LIOUVILLE EMPIRICAL-PATH LARGE DEVIATIONS AND INFORMATION PROJECTIONS FOR FINITE-HORIZON SINAI BILLIARDS

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ABSTRACT. The inducing process is treated as a graph-completed Markov-renewal process stopped at deterministic collision or physical time. The state retains singularity exposure, completed profiles, the next excursion and its pointed terminal prefix. An exact entropy-control representation gives one fixed good recurrent-recession rate and separate deterministic-clock contractions, without deleting finite-rate singular excursions or using a random speed.

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CONTENTS

1. Round-thirteen positive closure: graph-completed stopped renewal LDPs	1
1.1. The stopped Markov-renewal state	2
1.2. Exact entropy representation	2
1.3. The fixed recurrent and recession rates	3
1.4. Full collision-clock LDP	4
1.5. Physical-time stopping	4
Scope and verification status	5

1. ROUND-THIRTEEN POSITIVE CLOSURE: GRAPH-COMPLETED STOPPED RENEWAL LDPs

The singularity information is retained in the state and charged at its true large-deviation cost. It is never removed uniformly over an order- n entropy-control class. The terminal incomplete excursion is generated by the same controlled transition as the completed excursions, so there is no separate terminal likelihood and no double counting.

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1.1. The stopped Markov-renewal state. Let $\mathcal{A}$ be the countable inducing alphabet. A symbol a carries its source and target vertices, collision length $r(a) \geq 1$, physical roof $\tau(a) > 0$, and a graph-completed excursion $K_a : [0, r(a)] \rightarrow \bar{\Omega}_{\text{det}}$. The graph completion records the one-sided continuation, signed distance to every singularity chart, and the homogeneity label. Let $P(a, db)$ be the normalized Gibbs transition kernel.

For a deterministic collision horizon n put

$$S_m = \sum_{j=1}^m r(A_j), \quad N_n = \max\{m : S_m \leq n\}, \quad \alpha_n = n - S_{N_n}.$$

The stopped state Z_n^c contains

- (i) the transition occupation measure of $A_0, \dots, A_{N_n+1}$;
- (ii) the length-weighted empirical graph-completed profiles of all completed excursions;
- (iii) the next symbol A_{N_n+1} , its age α_n , and the pointed prefix $K_{A_{N_n+1}}|_{[0, \alpha_n]}$.

All masses are divided by the deterministic clock n .

Lemma 1.1 (Drift and compact containment for the actual renewal kernel). *There are $\eta > 0$, $0 < \rho < 1$, a finite set $C \subset \mathcal{A}$, and $b < \infty$ such that for $V(a) = e^{\eta(r(a)+\tau(a))}$,*

$$PV(a) \leq \rho V(a) + b1_C(a).$$

Moreover the graph-completed profile laws with $\int V d\nu \leq M$ are compact in the weighted local-Skorokhod topology.

Proof. The Young-tower return tail and bounded Gibbs distortion give an exponential moment for every branch leaving the finite connector set. Enlarging the connector set makes the tail contraction smaller than one and yields the drift inequality. A bound on V gives uniform clock tails. On a fixed clock level, finite horizon and homogeneity coordinates give equicontinuity on each regular segment, while the graph completion records every one-sided jump at a singularity. Diagonal Arzelà–Ascoli over path windows and the exponential tail yield compactness. No supremum over all entrance symbols is assumed; the drift controls the entrance dependence. $\square$

1.2. Exact entropy representation. A predictable control is a sequence of kernels $Q_j(\cdot | A_0, \dots, A_j)$ absolutely continuous with respect to $P(A_j, \cdot)$. It chooses A_{j+1} , including the symbol that contains the terminal residual.

Theorem 1.2 (Exact stopped-control variational formula). *For every bounded continuous F on the stopped state space,*

$$-\frac{1}{n} \log \mathbb{E} e^{-nF(Z_n^c)} = \inf_Q \mathbb{E}_Q \left[F(Z_n^{c,Q}) + \frac{1}{n} \sum_{j=0}^{N_n} D(Q_j(\cdot | \mathcal{F}_j) \| P(A_j, \cdot)) \right].$$

There is no independent terminal entropy term: the last summand is the likelihood of the branch whose prefix is observed at time n .

Proof. Apply the Gibbs variational formula successively to the conditional law of A_{j+1} until the stopping index is reached. Truncate at $N_n \wedge L$, use the ordinary finite-product chain rule for relative entropy, and let $L \rightarrow \infty$. Lemma 1.1 gives uniform integrability of the stopping index and its entropy costs. The terminal age and prefix are measurable functions of the same last branch, so the chain rule counts that branch exactly once. $\square$

1.3. The fixed recurrent and recession rates. Let $\mathcal{G}$ be the set of invariant edge measures $\eta(da, db) = \eta_1(da)q(a, db)$ with equal incoming and outgoing marginals and finite mean clock $\bar{r}(\eta) = \int r(b) d\eta$. Its recurrent entropy per collision clock is

$$\mathcal{R}_c(\eta) = \frac{\int D(q(a, \cdot) \| P(a, \cdot)) \eta_1(da)}{\bar{r}(\eta)}.$$

For a normalized graph-completed excursion profile ξ , define the actual one-excursion recession cost by

$$J_c(\xi) = \sup_{U \ni \xi} \lim_{L \rightarrow \infty} \inf_{\ell \geq L} \left[-\frac{1}{\ell} \log P \left(r(A_1) = \ell, \ell^{-1} K_{A_1} \in U \mid A_0 \in C \right) \right]^{\text{lsc}},$$

where connector changes alter the numerator by $O(1)$ and hence do not change the clock-normalized limit.

Theorem 1.3 (Joint recurrent–recession rate). *The collision-clock rate is the fixed lower-semicontinuous functional*

$$I^c(z) = \inf \left\{ (1 - \alpha) \mathcal{R}_c(\eta) + \alpha \int J_c(\xi) \zeta(d\xi) : \mathcal{C}_c(\eta, \alpha, \zeta, \rho) = z \right\}.$$

Here $0 \leq \alpha \leq 1$ is the fraction of collision clock carried by macroscopic excursions, ζ is their clock-weighted profile law, ρ is the pointed terminal residual law, and $\mathcal{C}_c$ is the explicit clock-weighted reconstruction map. The sublevels of I^c are compact. No singularity shield is discarded: singularity exposure is part of ξ and ρ and is charged by J_c .

Proof. The entropy bound and Lemma 1.1 give tightness of controlled edge occupations after separating a vanishing number of excursions whose individual lengths are of order n . The recurrent part converges to an invariant edge measure and the entropy chain rule gives $(1 - \alpha) \mathcal{R}_c$. Each macroscopic excursion is evaluated at its actual length; the definition of J_c gives the Gamma-liminf. Conversely, truncate η to a finite connector core, approximate it by a periodic Markov flow with the same vertex balance, and concatenate actual long excursions realizing finitely supported approximations of ζ . Connector cost and the pointed terminal prefix are $o(n)$ except when the prefix carries a positive clock fraction, in which case it is included in ρ and paid by the same one-excursion cost. Diagonal approximation gives

recovery. Clock exponential moments and graph-profile compactness make the sublevels compact. $\square$

1.4. Full collision-clock LDP.

Theorem 1.4 (Good stopped collision-clock LDP). *The states Z_n^c satisfy a full weak LDP with deterministic speed n and good rate I^c . The lower bound holds at every finite-rate state, not only at exposed points. Contracting the graph-completed profile to the physical collision path gives the physical empirical-path LDP; finite-rate states give zero mass to genuinely multivalued grazing continuations.*

Proof. Theorem 1.2 and compactness of bounded-cost controls give the Laplace upper bound by the weak-convergence method. The liminf is Theorem 1.3. Its recovery construction, inserted in the variational formula, gives the matching lower bound. Laplace and LDP forms are equivalent on the Polish compact-containment topology. The graph-to-physical map is continuous at every deterministic graph-completed path. A law charging a genuinely multivalued singular continuation has arbitrarily large one-excursion exposure cost and therefore infinite rate. $\square$

1.5. Physical-time stopping. Let $\Theta_m = \sum_{j=1}^m \tau(A_j)$ and stop at deterministic physical time T . The state Z_T^p retains the current excursion, the physical age, and the partial free flight. Define invariant controls with mean roof $\bar{\tau}(\eta)$ and recurrent cost

$$\mathcal{R}_p(\eta) = \frac{\int D(q\|P) d\eta_1}{\bar{\tau}(\eta)}.$$

The physical recession cost J_p is defined by normalizing actual one-excursion probabilities by physical duration.

Theorem 1.5 (Physical-clock LDP). *As $T \rightarrow \infty$, Z_T^p satisfies a full good LDP with speed T and rate*

$$I^p(z) = \inf \left\{ (1 - \alpha) \mathcal{R}_p(\eta) + \alpha \int J_p(\xi) d\zeta : \mathcal{C}_p(\eta, \alpha, \zeta, \rho) = z \right\}.$$

This is obtained from a second stopped-control representation and not by dividing a nonstationary collision-clock cost by one informal mean.

Proof. Repeat Theorem 1.2 with the stopping index $N(T) = \max\{m : \Theta_m \leq T\}$. Tight controlled occupation measures carry their roof coordinate, so recurrent entropy is normalized by the limiting roof occupation and macroscopic excursions by their actual physical duration. The same liminf and recovery arguments apply with the partial terminal flight included in ρ . This proves the rate and its compactness directly at deterministic speed T . $\square$

Theorem 1.6 (Round-thirteen A3 closure). *The Sinai inducing process has full good empirical-path LDPs at deterministic collision and physical clocks,*

including recurrent flow, macroscopic excursions, singularity exposure, and the pointed terminal residual in one exact state. Finite-rate singular events are retained rather than declared uniformly negligible under entropy controls.

Proof. Combine Theorems [1.2](#), [1.3](#), [1.4](#), and [1.5](#). □

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-thirteen dependency ledger. Repository source, proof-structure, direct-counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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